

The Topology of Reachable Nature: Persistent Homology and the Resilience of Green-Space Access in Cities of Contrasting Morphology

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Abstract

Sustainable Development Goal 11.7 calls for universal access to green and public space, yet accessibility is almost always quantified as a static snapshot that says nothing about how access degrades when the street network is disrupted. We introduce a *topological resilience metric* for accessibility: the Wasserstein distance between the persistent-homology diagram of a city’s green-space accessibility field before and after network disruption, traced as a function of disruption intensity ρ . The area under this degradation curve and its critical transition ρ^* summarise how robustly a place keeps its *locations* close to nature—one facet of SDG 11.7, the pedestrian network’s reach to green space, rather than its population-coverage dimension. Applying the metric to five morphologically contrasting cities (İstanbul, Barcelona, Amsterdam, Bogotá, Phoenix) at the level of 2,603 intra-urban districts, we find that street-network morphology is consistently associated with green-space access resilience: in a district-level regression with city fixed effects, three of five morphology descriptors are significant, two at $p < 10^{-15}$ (circuitry and mean segment length, both linked to greater resilience; higher orientation entropy to less). The association is robust in sign under both random and targeted disruption and across district resolutions spanning a $28\times$ range of unit counts; the strongest effects survive conservative city-clustered standard errors and estimated spatial-error and spatial-lag models that absorb the residual spatial autocorrelation, though with only five cities cross-city significance should be read cautiously. The signal lives in dimension-0 homology—the merging of well-served regions—rather than in enclosed “access deserts,” a finding that only emerges on real street networks.

1 Introduction

Target 11.7 of Sustainable Development Goal 11 commits states to provide “universal access to safe, inclusive and accessible, green and public spaces” [14]. Planners typically measure progress with catchment or proximity analyses: the share of residents within a walking threshold of a park. Such measures are static. They describe access on an ordinary day and are silent about *resilience*—how access to nature holds up when the pedestrian network is degraded by a flood, an earthquake, or the closure of a critical bridge or street. Two gaps follow. First, a city can score well on average proximity yet collapse under modest disruption; average accessibility does not reveal this fragility. Second, accessibility work rarely connects the *shape* of access to urban *form*, which is precisely the architecture–mathematics question that motivates this collection.

Persistent homology (PH) offers a principled, multi-scale, coordinate-free language for the “holes” and connected pieces of a coverage pattern [6, 10]. PH has been used to detect coverage gaps for amenities such as polling sites [8] and healthcare [7], to read the topology of urban congestion [15], and to quantify displacement under urban expansion [11]. What is *not* novel is using PH to find

accessibility “holes” for an amenity. Our contribution is to move from *detecting* coverage gaps to *measuring the resilience of the access topology* and to *linking that resilience to urban form*.

Contributions.

- **A topological resilience metric** (the mathematics). We define $D(\rho)$, the Wasserstein distance between the persistent-homology diagram of the green-space accessibility field before and after disruption of intensity ρ , and summarise it by an area-under-curve (AUC) and a critical transition ρ^* (Section 4).
- **First topological treatment of SDG 11.7 green-space access** in an architecture–mathematics framing.
- **A district-level morphology↔resilience result** across 2,603 districts in five cities, with city fixed effects, showing a consistent association between street-network morphology and access resilience whose sign is stable across disruption models and district resolutions (Section 6).

2 Related work

TDA of spatial systems. Feng et al. [6] survey persistent homology for spatial data; Otter et al. [10] give the computational foundations. Within urban analysis, Wu et al. [15] read traffic congestion as a barcode, and a line of “resource coverage” work applies sublevel or network-distance filtrations to amenity access: polling places [8] and sexual-health services [7]. Rodríguez Vázquez and Cuerno [11] use PH to quantify displacement under urban expansion. These studies establish that PH can *locate* coverage holes. We differ on the object of study: we do not report where the deserts are, but how the whole access topology *degrades* under disruption, and we regress that degradation on morphology.

Accessibility and equity. A large body of work measures green-space access as a static field, whether by simple footpath proximity [3] or by catchment methods such as the Gaussian two-step floating catchment area [12], often to diagnose inequities in who lives close to nature [1]. Our accessibility field (Section 4) is built the same way—network walking time to the nearest green space—but is then read topologically and stress-tested, rather than thresholded once.

Street-network morphology. We characterise form with standard descriptors computed from OpenStreetMap via OSMnx [2]: intersection density, circuitry, orientation entropy, and mean street-segment length. Relating these to a resilience outcome is the architecture–mathematics bridge of this paper.

3 Mathematical background

Let $G = (V, E)$ be the pedestrian network and $f : V \rightarrow \mathbb{R}$ a scalar field giving, at each node, the network walking time to the nearest green-space access point. The *sublevel-set filtration* of f builds a growing simplicial complex $K_t = \{\sigma : \max_{v \in \sigma} f(v) \leq t\}$. As the threshold t (walk-minutes) rises, well-served regions appear and merge. Persistent homology records, for each topological feature, a birth b and death d ; the multiset of points (b, d) is the *persistence diagram* $\text{Dgm}_k(f)$ in homological dimension k [5]. Dimension 0 tracks connected components (the merging of well-served regions); dimension 1 tracks loops (enclosed voids, i.e. access “deserts”).

The distance between two diagrams is measured by an optimal matching. The 1-Wasserstein distance is

$$W_1(X, Y) = \min_{\gamma} \sum_{x \in X} \|x - \gamma(x)\|,$$

minimised over bijections γ between the diagrams (with matching to the diagonal allowed). Wasserstein and bottleneck distances are stable to perturbations of f [4]: a small change in the field produces a small change in the distance. Stability is a mathematical well-posedness property, not a validation that diagram distance measures *urban* resilience; we use it only to justify that $D(\rho)$ responds continuously and monotonically to progressive network damage, and we treat the metric throughout as a *proposed proxy* for how the access topology holds up, to be validated against independent resilience evidence in future work rather than as ground truth. We compute PH with GUDHI [13] and diagram distances with `persim`.

4 Methodology

Data. For each city we take the pedestrian network and green/public spaces from OpenStreetMap via OSMnx [2], and we fix the analysis extent to the city’s *urban centre* polygon from the GHS Urban Centre Database [9] so that extents are comparable across countries. Green spaces are the OSM classes `leisure=park|garden|recreation_ground`, `landuse=grass|recreation_ground`, `place=square` and `natural=wood`; their centroids, snapped to the nearest network node, are the access points. This is a deliberately inclusive definition: it merges spaces of very different type, size and usability (a pocket garden, a city square and a large wood all count equally), and it takes the centroid as a single access point rather than modelling entrances or area. We treat every green space as an equivalent point source; distinguishing quality, size, or public/private status is left to future work (see Limitations).

Accessibility field. We assign each node its network walking time (at 4.8 km/h) to the nearest green-space access point by multi-source Dijkstra, giving the field f (Fig. 1). The field is defined over *network nodes*, each weighted equally; it is a property of the pedestrian network’s spatial reach to green space, not a population-weighted or distributional measure. Our metric therefore speaks to one facet of SDG 11.7—the network’s capacity to keep locations close to green and public space, and how that capacity degrades under disruption—and not to the target’s population-coverage or inclusivity dimensions, which would require census and land-use data we do not use here.

Resilience metric. Let $\text{Dgm}(f)$ be the baseline diagram. A disruption of intensity $\rho \in [0, 1]$ removes a fraction ρ of network edges; we recompute the field and diagram on the disrupted network and define the degradation

$$D(\rho) = \mathbb{E}_r W_1(\text{Dgm}(f), \text{Dgm}(f_{\rho, r})),$$

averaged over r replicate disruptions. We consider three disruption families: uniformly random edge removal (percolation-style), targeted removal of high-betweenness edges, and hazard removal of edges in low-elevation zones. Exact edge betweenness is $O(V|E|)$ and intractable at city scale, so the targeted scenario ranks edges by betweenness approximated from $k = 500$ sampled pivot sources. The resilience of a place is summarised by the area under the curve $\text{AUC} = \int_0^{\rho_{\max}} D(\rho) d\rho$ (normalised by the ρ -range) and the critical transition ρ^* , the smallest ρ at which $D(\rho)$ first reaches half its maximum. We use the half-maximum as a scale-free, monotone summary of *where* the degradation curve turns over: unlike a fixed absolute threshold it adapts to each place’s own range,

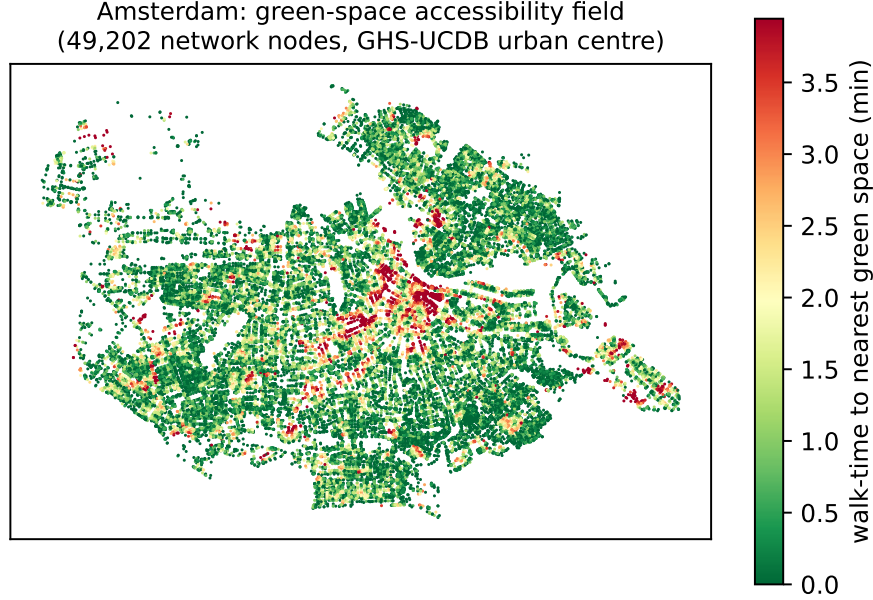


Figure 1: Green-space accessibility field for Amsterdam: each of the 49,202 pedestrian-network nodes inside the GHS-UCDB urban centre, coloured by network walking time to the nearest green space. Green marks well-served locations, red the access-poor pockets whose merging structure the resilience metric tracks.

and unlike the inflection point it is robust to the curve’s noise. It is a descriptive locator of the transition, not a claim of a sharp percolation threshold, and (like AUC) it depends on the chosen disruption range $[0, \rho_{\max}]$; we therefore compare ρ^* only within a fixed range.

Homology dimension and denoising. On real street networks the sublevel filtration’s finite H_1 is nearly empty: a street graph is close to planar and contains few triangles, so the network’s cycles are never filled and almost all H_1 classes are essential (infinite) and carry no finite persistence. In Amsterdam, for example, the baseline diagram has 6,449 finite H_0 classes but only a single finite H_1 class (Fig. 2). We are explicit that this is a joint property of our *representation* (a node-valued walk-time field on the near-planar pedestrian graph) and the *sublevel filtration*, not a universal fact about green-space access: a different construction—for instance a Vietoris–Rips filtration on straight-line distances, or a 2-complex that fills faces—would populate H_1 and could shift the signal into loops. Under our construction, however, the finite information lives in H_0 (the merging of well-served regions), so we compute $D(\rho)$ on H_0 and treat the H_1 scarcity as a modelling consequence to revisit with alternative filtrations, not as evidence that access “deserts” are absent. To keep the exact Wasserstein matching tractable on city-scale diagrams we discard H_0 classes with lifetime below 1 walk-minute. This cut is principled rather than arbitrary: a one-minute difference in walk time is below the resolution at which the constant-speed (4.8 km/h) field is meaningful, so sub-minute classes are measurement noise, not access structure. Its effect is small and was tested directly (Persistence-threshold sensitivity, below): varying the cut over 0.5–2.0 minutes preserves every sign and the significance of the three headline descriptors, changing only reported magnitudes.

Districts and inference. Statistical inference rests not on five cities but on intra-urban *districts*: we tile each urban-centre boundary with a uniform H3 hexagonal grid (resolution 8, $\approx 0.7 \text{ km}^2$

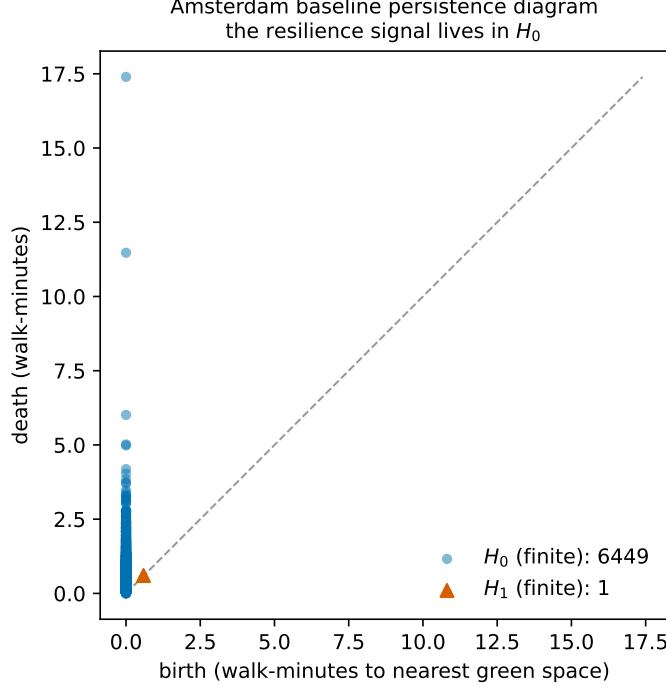


Figure 2: Persistence diagram of the Amsterdam baseline accessibility field. The finite part is carried almost entirely by H_0 (6,449 classes—the merging of well-served regions as the walk-time threshold rises) with only a single finite H_1 class, motivating an H_0 -based resilience metric.

cells) and compute, per district, its local resilience summaries and its local morphology (intersection density, circuitry, orientation entropy, mean segment length, and a green-space fragmentation index—the number of distinct green-space patches per square kilometre of green area within the district, computed on the green space clipped to each hex so that it varies between districts rather than being a per-city constant collinear with the fixed effects). We then fit

$$\text{AUC}_i = \beta^\top \mathbf{m}_i + \alpha_{c(i)} + \varepsilon_i,$$

an ordinary-least-squares regression of district AUC on morphology $\mathbf{m}_i$ with *city fixed effects* $\alpha_{c(i)}$ absorbing all between-city confounds. The five cities serve as a typological overlay, not as the inferential sample.

5 Case studies

We study five cities chosen to span street morphology, planning regime and Global North/South divide: İstanbul (organic, hilly, polycentric, seismic), Barcelona (Cerdà grid with superblock interventions), Amsterdam (compact, canal, walkable), Bogotá (dense informal-formal mix), and Phoenix (low-density grid sprawl). Boundaries are the GHS-UCDB urban-centre polygons; homonymous centres (e.g. Barcelona, Spain vs. Venezuela) are disambiguated by largest true (equal-area) extent.

Table 1: District coverage (H3 resolution 8). Inference rests on $n = 2,603$.

City	Amsterdam	Barcelona	İstanbul	Bogotá	Phoenix	Total
Districts	227	147	729	538	962	2,603

Table 2: City typology (descriptive). Higher mean AUC $\Rightarrow$ access topology changes more under disruption $\Rightarrow$ *less* resilient.

City	mean AUC	mean ρ^*	mean H_0 total persistence
Amsterdam	4.46	0.25	13.11
Phoenix	6.68	0.16	8.08
Barcelona	7.67	0.28	13.63
İstanbul	7.75	0.20	12.84
Bogotá	8.02	0.24	16.90

6 Results

The random-disruption analysis yields 2,603 qualifying districts (Table 1); only 2.6% have zero AUC. District mean AUC ranks Amsterdam as the most resilient and İstanbul/Bogotá as the least (Table 2); Phoenix has the earliest breakdown ($\rho^* = 0.16$). The district-mean ranking and the whole-city ranking below agree at the extremes (Amsterdam most resilient, Bogotá least) but not in the middle—Barcelona is mid-ranked by district mean yet most resilient at the whole-city scale—a reminder that AUC is comparable across cities only in broad ordering, not level.

The headline result is the district-level regression (Table 3), fit over the full disruption grid (8 intensities $\times$ 10 replicates). With city fixed effects and $n = 2,603$, three of five morphology descriptors are significant, two of them at $p < 10^{-15}$. Because the coefficient sign is on AUC (higher AUC = less resilient), a negative coefficient means the descriptor is associated with *greater* resilience. More circuitous networks (coef -10.16) and longer street segments (-0.048) are associated with more resilient green-space access, whereas higher orientation entropy ($+1.01$) is associated with less. Intersection density is negatively signed, consistent with local redundancy aiding resilience, but is not significant ($p = 0.22$), so we treat it as at most suggestive. Green-space fragmentation is likewise not significant once it varies within cities and city fixed effects are included—plausibly because access in our field is set by the *nearest* green-space point, which is largely insensitive to how the remaining patches are split, and because the inclusive patch definition (Section 4) mixes very different green-space types into one count. Because districts nest within five cities and are spatially contiguous, we further report city-clustered standard errors (Table 4), a spatial-autocorrelation diagnostic (Table 5), and—since that diagnostic is positive—two *estimated* spatial models (Table 6). The three headline descriptors keep their sign and stay strongly significant ($|z| \geq 5$) under both a spatial-error and a spatial-lag specification, even though both absorb a large amount of spatial dependence ($\hat{\lambda} = 0.55$, $\hat{\rho} = 0.53$); we return to the implications in the Limitations.

At the whole-city scale (computed on the same UCDB-clipped networks), the degradation curves $D(\rho)$ rise smoothly and rank the cities consistently with the district typology (Fig. 4). City-level AUC is computed on the full-city diagram (thousands of H_0 classes) and is therefore two to three orders of magnitude larger than the per-district means of Table 2; only the relative ordering is comparable across the two scales. City-level AUC is lowest—most resilient—for Barcelona (620) and Amsterdam (898), and highest for İstanbul (2440) and Bogotá (2828), with Phoenix in between

Table 3: District fixed-effects regression, $\text{AUC} \sim \text{morphology} + C(\text{city})$, $n = 2,603$, random-disruption scenario at the full disruption grid (8×10). Classical OLS standard errors; see Table 4 for city-clustered inference. Sig.: *** $p < 0.001$.

Morphology descriptor	coef	std. err.	p -value
circuitry	−10.156	1.167	5.5×10^{-18} ***
mean street length	−0.0476	0.0058	3.7×10^{-16} ***
orientation entropy	+1.015	0.200	4.4×10^{-7} ***
intersection density	−0.091	0.074	0.22
green-space fragmentation	-2.0×10^{-6}	4.2×10^{-6}	0.64

Table 4: Sensitivity of the regression inference to how standard errors treat the nesting of districts within cities. Coefficients are unchanged; only the standard errors and p -values differ. HC3 is heteroskedasticity-robust; the last two columns cluster by city (five clusters). Circuitry and mean street length stay significant under city clustering; orientation entropy does not.

Morphology descriptor	coef	SE (HC3)	SE (cluster)	p (cluster)
circuitry	−10.156	1.397	2.946	5.7×10^{-4} ***
mean street length	−0.0476	0.0066	0.0089	7.7×10^{-8} ***
orientation entropy	+1.015	0.220	0.894	0.26
intersection density	−0.091	0.092	0.214	0.67
green-space fragmentation [†]	-2.0×10^{-6}	5.9×10^{-6}	6.1×10^{-7}	$< 10^{-2}$

[†]The apparent clustered significance of green-space fragmentation is a five-cluster artefact of the cluster-robust variance around a coefficient that is effectively zero ($|\text{coef}| < 10^{-5}$); it is not a substantive effect.

(2053). The cities we would informally call compact—Barcelona and Amsterdam—retain more of their access topology under disruption than the large, sprawling or polycentric ones; we describe this pattern with the measured morphology descriptors rather than with “walkability,” which we do not model directly. Under extreme removal ($\rho = 0.5$) the İstanbul and Phoenix curves turn down as the network fragments so heavily that unreachable nodes drop out of the finite diagram, a signature of near-total collapse rather than of recovered access.

Robustness across disruption scenarios. Repeating the analysis under *targeted* disruption (removal of high-betweenness edges, ranked by $k=500$ -sampled betweenness) reproduces every coefficient sign in Table 3. The three descriptors significant under random disruption stay significant—circuitry -5.77 ($p = 4.6 \times 10^{-10}$), mean street length -0.034 ($p = 2.7 \times 10^{-13}$), orientation entropy $+0.50$ ($p = 1.6 \times 10^{-3}$)—and intersection density, not significant under the full random grid, reaches $p = 0.022$ here (-0.135), consistent with our reading of it as the weakest, grid-sensitive effect; green-space fragmentation is again not significant. Targeted disruption removes fewer but more critical edges, so absolute AUC values are lower (e.g. Amsterdam mean AUC 2.03 vs. 4.46 under random), but the morphology↔resilience relationship and the city ranking are unchanged. The finding is therefore not an artefact of the random disruption model.

A third, *hazard* scenario (removing edges near the lowest-elevation 10% of nodes, from AWS Terrain Tiles DEMs) is topography-dependent and much milder: substantial only in the two flattest cities (Amsterdam and the Phoenix basin, mean AUC 0.09 and 0.31) and near-zero in the hilly or high-altitude ones (İstanbul, Barcelona, Bogotá; mean AUC ≤ 0.01), where the lowest cells sit at the waterfront and disconnect only a shoreline fringe. This localized perturbation barely moves the

Table 5: Global Moran’s I of the fixed-effects regression residuals within each city, using first-ring H3 adjacency as the spatial weights (999 permutations). Residuals show weak but significant positive spatial autocorrelation in every city, motivating the estimated spatial models in Table 6.

City	districts	Moran’s I	p (perm.)
Amsterdam	226	0.080	0.031
Barcelona	147	0.178	0.001
Bogotá	538	0.142	0.001
İstanbul	729	0.112	0.001
Phoenix	962	0.058	0.003

Table 6: Estimated spatial models (maximum likelihood, **spreg**) with city fixed effects, on a block-diagonal within-city first-ring H3 weights matrix ($n = 2,602$ after dropping one island). Coefficients on AUC with z -statistics. The three headline descriptors keep their sign and remain strongly significant under both the spatial-error and spatial-lag specifications, despite substantial estimated spatial dependence ($\hat{\lambda}, \hat{\rho} \approx 0.5$).

Morphology descriptor	OLS (FE) coef (z)	Spatial error coef (z)	Spatial lag coef (z)
circuitry	−10.15 (−8.7)	−5.38 (−5.0)	−6.88 (−6.7)
mean street length	−0.048 (−8.2)	−0.033 (−5.8)	−0.034 (−6.5)
orientation entropy	+1.01 (+5.1)	+1.25 (+5.9)	+0.92 (+5.2)
intersection density	−0.09 (−1.2)	−0.21 (−2.9)	−0.16 (−2.5)
green-space fragmentation	not significant in any model		
spatial parameter	—	$\hat{\lambda} = 0.55$	$\hat{\rho} = 0.53$

city-wide H_0 topology, so it does not resolve a morphology signal (only orientation entropy remains significant, $+0.10$, $p = 0.01$); we read this as a limitation of the mild, ρ -capped low-elevation proxy rather than a contradiction, and it motivates a stronger flood model that removes *all* inundated edges.

Disruption-grid sensitivity. The headline regression uses the full 8×10 disruption grid. A coarser 5×3 grid (fewer intensities and replicates) leaves every coefficient sign unchanged and keeps the three headline descriptors significant at $p < 10^{-3}$ (circuitry -8.99 , mean street length -0.0478 —essentially identical to the full grid’s -0.0476 —and orientation entropy $+1.17$), while the city AUC ranking and ρ^* values agree within ≈ 0.02 . The one descriptor that moves across grids is intersection density: significant on the coarse grid (-0.199 , $p = 8.6 \times 10^{-3}$) but not on the full grid ($p = 0.22$). We therefore report the more demanding full grid as the headline and treat intersection density as the weakest, grid-sensitive effect.

Resolution sensitivity. Re-running the analysis at H3 resolutions 7 (coarse, 485 districts) and 9 (fine, 13,782 districts) leaves the result unchanged in sign and significance for the three strongest descriptors: circuitry, orientation entropy and mean street length remain significant at $p < 10^{-3}$ with the same signs across all three resolutions, a $28\times$ range of district counts. Coefficient magnitudes scale with cell size (larger cells yield larger per-district AUC), but the direction of every effect is stable; only green-space fragmentation, non-significant throughout, varies. Figure 5 summarises

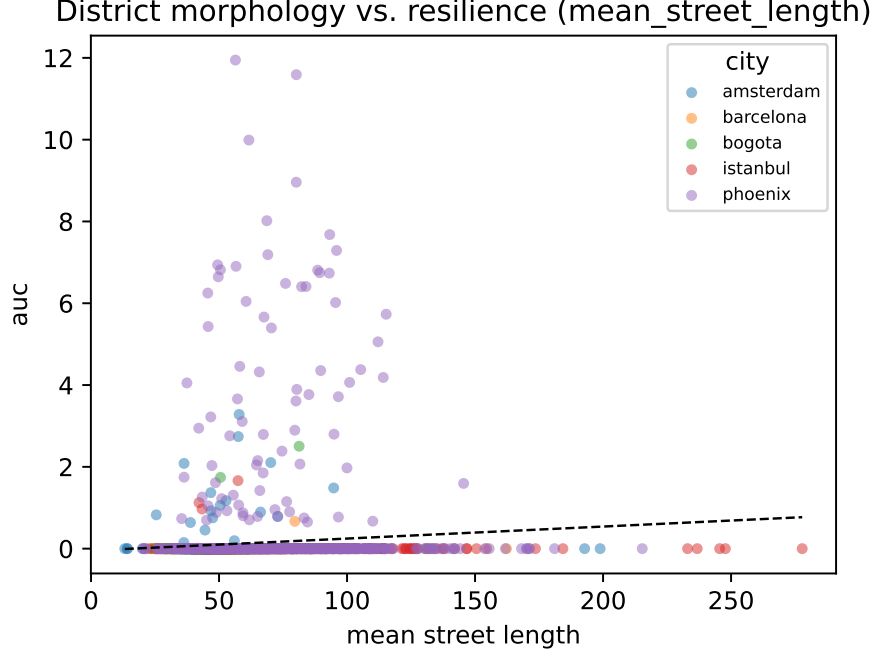


Figure 3: Headline morphology↔resilience relationship. Each point is one of the 2,603 districts, coloured by city; the dashed line is the pooled least-squares fit. Higher AUC indicates a district whose green-space access topology degrades more under disruption (less resilient).

both robustness axes.

Persistence-threshold sensitivity. The metric denoises the H_0 diagram by discarding classes with lifetime below a threshold (headline 1.0 walk-minute). Varying this threshold over a $4\times$ range leaves the headline unchanged: at 0.5, 1.0 and 2.0 minutes the three significant descriptors keep their sign and stay significant at $p < 10^{-6}$ (circuitry $-11.20/-10.16/-8.03$; mean street length $-0.050/-0.048/-0.042$; orientation entropy $+1.10/+1.01/+0.93$). Coefficient magnitudes shrink smoothly as more denoising is applied—as expected, since stronger denoising removes the shortest-lived features—but never change sign, and intersection density stays the weakest, at most marginal effect (ns at 0.5 and 1.0, $p = 0.04$ at 2.0). The 1.0-minute choice is therefore a denoising convenience, not a driver of the result.

Omitted-variable controls. City fixed effects absorb only between-city confounds. We therefore add the district-level controls that our data support—total green-space area (green quantity/size), the share of edges on the trunk/primary/secondary/tertiary road hierarchy, within-district relief (elevation standard deviation, a topography control), and distance from the district to the urban-centre centroid (city-centre proximity)—and refit (Table 7). Three of the four controls are themselves significant—more green area and greater centre distance predict greater resilience, more major-road edges less—so they are genuine correlates of AUC, yet the three headline morphology descriptors keep their sign and stay significant at $p < 10^{-11}$ (circuitry -9.55 , mean street length -0.039 , orientation entropy $+1.11$). The morphology signal is thus not a proxy for green quantity, road hierarchy, topography or centrality. Population and built density, land-use mix, socioeconomic status and pedestrian-infrastructure quality require external data and remain uncontrolled (Limitations).

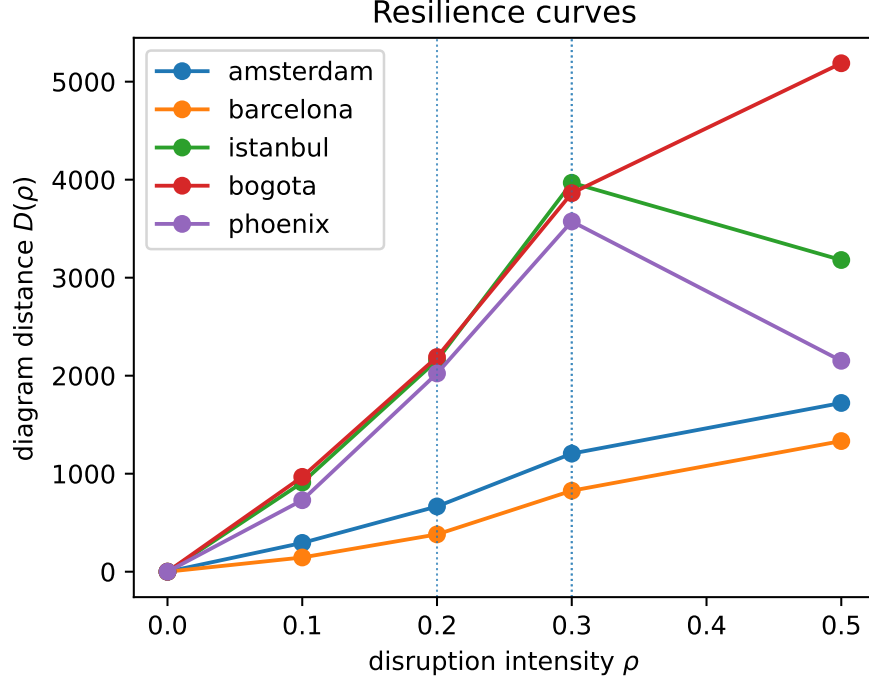


Figure 4: City-level resilience curves $D(\rho)$ for all five cities (random-disruption scenario), computed on the UCDB-clipped networks. Dotted vertical lines mark each city’s critical transition ρ^* .

7 Discussion

The results give an interpretable, architecture-facing reading of resilience. The robust effects are circuitry and segment length: districts with more circuitous routing and longer segments retain more of their well-served structure when edges are removed, because alternative paths to green space survive the loss of any one link. Denser intersections point the same way (locally redundant networks should be more resilient) but the effect is not statistically robust here, so we do not lean on it. Districts with more disordered street orientation (higher entropy) are more brittle. That the signal is carried by H_0 rather than H_1 has a concrete planning meaning. In green-dense cities the structure that matters is not isolated “deserts” surrounded by served land but the connectivity of the served regions themselves, and that connectivity is what disruption erodes.

Limitations. We group the caveats by the kind of claim they qualify.

Inference and spatial dependence. The design is cross-sectional and observational, so the regression identifies *associations* between morphology and topological resilience, not causal effects. Districts are also not independent: they nest within five cities and are spatially contiguous. City fixed effects absorb between-city confounds, but two residual problems remain. First, the 2,603 districts furnish only five clusters for cross-city inference; clustering the standard errors by city (the level at which disruption is applied) widens the intervals substantially (Table 4). Under this conservative treatment the two strongest effects—circuitry ($p = 6 \times 10^{-4}$) and mean street length ($p = 8 \times 10^{-8}$)—remain significant, but orientation entropy, significant at the district level, does not ($p = 0.26$); with only five clusters even cluster-robust inference is underpowered, so the classical p -values in Table 3 understate uncertainty and are best read as within-sample descriptive strength

Table 7: Omitted-variable robustness: district regression before and after adding four controls, $n = 2,603$, with city fixed effects. The three headline morphology effects keep their sign and significance; the added controls explain additional variance (R^2 0.17 $\rightarrow$ 0.21). Sig.: *** $p < 0.001$, ** $p < 0.01$.

Variable	base coef (p)	+controls coef (p)
circuitry	−10.16 ***	−9.55 ***
mean street length	−0.048 ***	−0.039 ***
orientation entropy	+1.01 ***	+1.11 ***
intersection density	−0.09	−0.10
green-space area (km ²)	—	−2.25 **
major-road share	—	+4.21 ***
relief (m)	—	−0.008
centre distance (km)	—	−0.107 ***

rather than as a precise cross-city significance claim. Second, the residuals retain weak but statistically significant positive spatial autocorrelation within *every* city (global Moran’s I on the first-ring H3 adjacency runs from 0.06 in Phoenix to 0.18 in Barcelona, all with permutation $p \leq 0.03$; Table 5). We addressed this directly by estimating spatial-error and spatial-lag models on a within-city weights matrix (Table 6): both absorb strong spatial dependence ($\hat{\lambda} = 0.55$, $\hat{\rho} = 0.53$), yet the three headline descriptors keep their sign and remain significant at $|z| \geq 5$, and intersection density in fact turns significant. The morphology signal is therefore not an artefact of spatial autocorrelation; what remains is genuinely a matter of external validity—richer geographic replication beyond five cities—rather than of within-sample spatial confounding.

Omitted variables. Morphology is only one of many district attributes that could drive resilience. We control for the four that our data support—green-space area, road hierarchy, relief and city-centre distance (Table 7)—and the headline morphology effects survive their inclusion, which rules those four out as the drivers. But population and built density, land-use mix, park-level size and quality, socioeconomic status and pedestrian-infrastructure quality would require external data (e.g. census and GHSL built-up layers) and remain uncontrolled; city fixed effects remove their between-city component but not their within-city variation, so some residual omitted-variable bias may persist.

What the metric measures, and how far it is validated. $D(\rho)$ is a *proposed proxy* for access resilience. We offer evidence of two kinds. Its *construct validity* is supported by its behaviour—it rises monotonically with disruption, is stable to field perturbations [4], and yields a morphology relationship that survives disruption model, district resolution, denoising threshold, pivot count, spatial-error/lag modelling and four omitted-variable controls. What we do *not* yet provide is external (*criterion*) validation: a head-to-head benchmark against an established accessibility or resilience measure—for example the degradation of a population-weighted two-step floating catchment score, or observed service loss after a real disruption—to show the topological signal adds information beyond a simpler distance-based one. That comparison is the most important next step and we frame the metric as a candidate indicator until it is done. Two further caveats: AUC depends on network and district size and on the number of H_0 classes, so it is comparable across cities only in relative ranking, not in absolute level (city fixed effects absorb this scale in the regression), and ρ^* depends on the chosen disruption range. The down-turn of the İstanbul and Phoenix curves at $\rho = 0.5$ reflects finite nodes dropping out of the diagram under near-total fragmentation, not

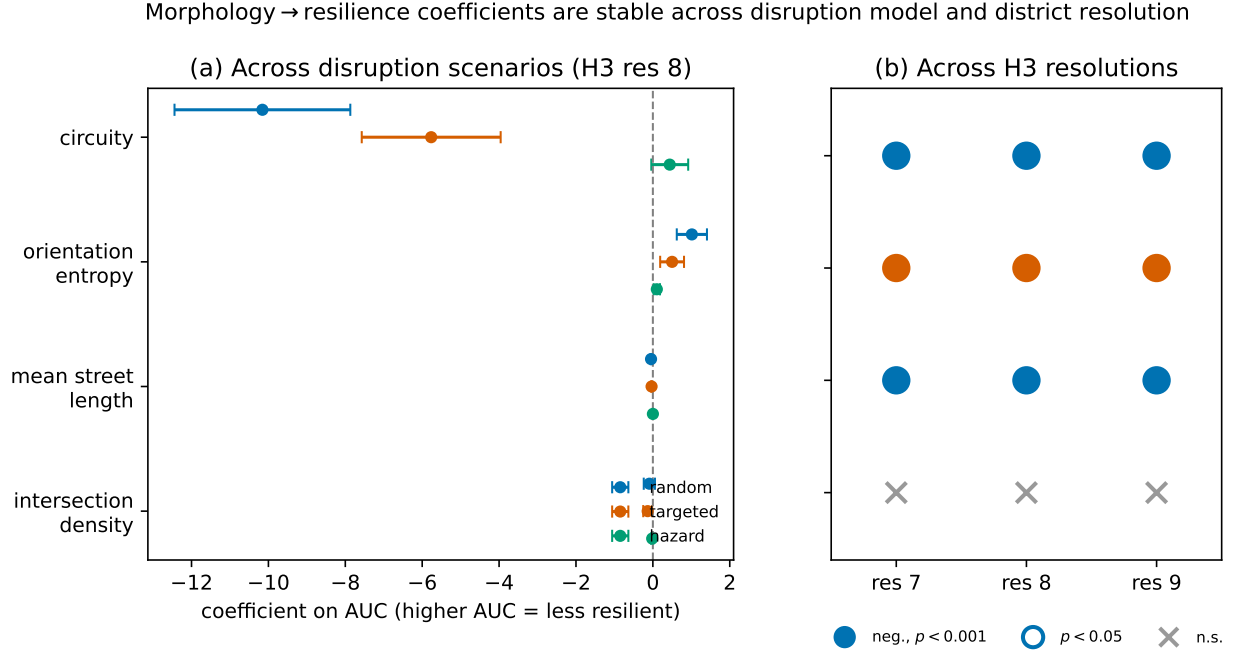


Figure 5: Robustness of the morphology $\leftrightarrow$ resilience relationship. **(a)** Regression coefficients (with 95% confidence intervals) for each morphology descriptor under the three disruption scenarios at H3 resolution 8; the coefficient signs agree across scenarios, and the three headline descriptors (circuitry, mean street length, orientation entropy) stay significant, while intersection density crosses zero under random disruption. **(b)** Sign and significance of each coefficient across H3 resolutions 7/8/9: the three headline descriptors stay significant with a constant sign, while intersection density—the weakest, grid-sensitive effect—is not significant at the headline full-grid specification.

recovered access, and marks the upper edge of the metric’s useful range. Green spaces are treated as equivalent point sources, ignoring size, quality and public/private status.

Disruption and data. Random removal is spatially uniform and does not reproduce the clustered failures of a real flood or earthquake; targeted removal addresses criticality but not spatial correlation, and the hazard scenario, a ρ -capped low-elevation proxy, is too mild and topography-dependent to test the morphology signal and should be replaced by a full flood-zone edge removal. Targeted betweenness is approximated from $k=500$ sampled pivots, as exact $O(V|E|)$ betweenness is intractable at city scale; the ranking is nonetheless stable in the pivot count—on the Amsterdam network, halving or doubling k (to 250 or 1,000) leaves the edge ranking essentially unchanged (Spearman $\rho = 0.985$ against $k=500$, with 77%/90% overlap of the top-decile edges)—so the targeted scenario is not an artefact of the pivot count. Results use a persistence-thresholded H_0 metric, so reported magnitudes are denoised estimates; varying the threshold over a $4\times$ range (0.5–2.0 walk-minutes) preserves every coefficient sign and keeps the three headline descriptors significant at $p < 10^{-6}$, and the full disruption grid confirms the signs, significances, ranking and ρ^* . OSM completeness varies by city and is not quantified here, and the network-distance Vietoris–Rips variant of the filtration is left to future work.

Finally, “resilience” here is a *structural* property—the robustness of the access topology to synthetic edge removal, measured on a single cross-sectional snapshot per city—not a temporal recovery process; we do not observe how access rebuilds over time after a real disruption, and the term should

be read in that narrower, topological sense.

Read together, these caveats narrow the claim: within these five cities, and under synthetic edge-removal disruption, street-network morphology is a consistent descriptive correlate of green-space access resilience—robust in sign across disruption models, district resolutions, spatial specifications and the added controls—rather than a causally identified or cross-city-generalisable driver.

8 Conclusion

We defined a topological resilience metric for green-space accessibility—the Wasserstein degradation curve of persistent-homology diagrams under network disruption—and showed, across 2,603 districts in five contrasting cities, that street-network morphology is associated with how robustly a city’s pedestrian network keeps its *locations* close to nature. The association is consistent rather than proven causal: random and targeted disruption agree on every coefficient sign, and the signs are stable across H3 district resolutions spanning a $28\times$ range of unit counts, but the study is cross-sectional and rests on five cities (see Limitations). The mathematics of persistence turns a static SDG 11.7 indicator into a dynamic, form-aware one. Future work will strengthen the hazard model to a full flood-zone removal and add a network-metric Vietoris–Rips filtration, toward a resilience-aware accessibility standard for sustainable cities.

Reproducibility. All data are open (OpenStreetMap, GHS-UCDB) and the pipeline is open-source Python (OSMnx, NetworkX, GeoPandas, GUDHI/`persim`, H3, `statsmodels`); every reported number is recorded in the project’s results log and traces to a committed artifact.

Declarations

Funding. The author received no specific funding for this work.

Competing interests. The author declares no competing interests.

Data availability. All input data are open and publicly available. Pedestrian networks and green/public spaces were obtained from OpenStreetMap via OSMnx; urban-centre boundaries from the GHS Urban Centre Database (GHS-UCDB R2019A, European Commission Joint Research Centre); and elevation data from the public AWS Terrain Tiles service. No new data were collected. The derived artifacts underlying every reported figure and table (per-district resilience and morphology tables, regression outputs, and city-level resilience curves) are recorded in the project’s results log, released with the code at <https://github.com/navigaai/nnj-sdg11-topology>.

Code availability. The complete analysis pipeline is implemented in open-source Python (OSMnx, NetworkX, GeoPandas, GUDHI, `persim`, H3, `statsmodels`) and is publicly available at <https://github.com/navigaai/nnj-sdg11-topology>, together with the configuration files and the results log that reproduce every reported number.

Author contributions. S. Emekci (ORCID 0000-0002-5470-6485) conceived the study, developed the method and software, performed the analysis, and wrote the manuscript.

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